Assignment10 – Natural Language

Given: June 24   Due: June 29

Problem 10.1 (Ambiguity)

1. Explain the concept of *ambiguity* of *natural languages*.
2. Give two examples of different kinds of *ambiguity* and explain the *readings*.

Problem 10.2 (Language Identification)

You are given an English, a German, a Spanish, and a French text corpus of considerable size, and you want to build a language identification algorithm A for the EU administration. Concretely A takes a string as input and classifies it into one of the four languages $\ell^* \in \{English, German, Spanish, French\}$. The prior probability distribution for the strings being English/German/Spanish/French, is $\langle 0.4, 0.2, 0.15, 0.15\rangle$.

How would you proceed to build algorithm A? Specify the general steps and give/derive the formula for computing ℓ given a string $\mathbf{c}_{1:N}$.

Problem 10.3 (Language Models)

1. How can we obtain a *trigram* model for a *natural language*?
   Explain the *probability distribution* involved.
2. Explain informally how we can use *trigram* models to identify the language of a document D.
3. Explain briefly what *named entity recognition* is.

Problem 10.4 (Part-of-Speech Tagging)

Consider the following corpus

A dog runs. The man sees the dog. The man shouts.

and the following POS tags: D (determiner) N (noun), V (verb), E (end of sentence).

1. Annotate the corpus with POS tags.
2. Compute the transition and the sensor model from the corpus.
   Note: To do so, you have to make reasonable choices for the border states (first and last tag of the text).
3. Now consider the sentence

   The man runs.

   Explain informally how you can use the HMM from above to assign POS tags to that sentence.